Dynamical System: the pendolum with oscillating suspension point

Assignment :

For the period shift map of the pendulum with oscillating suspension point in the parameter space (α, k) :

a. plot the first few instability tongues of the lower fixed point $z_- = (0, 0)$.

b. plot the stability region (regions ?) of the upper fixed point $z_+ = (\pi, 0)$.

THE MODEL:

Consider a pendolum with oscillating suspension point 2Pi-periodically.

After time rescaling the equation of motion is:

x''=-(α^2 - k Cos[t]) Sin[x] with initial conditions x(0)=x0, x'(0)=v0 ,

where k=a/l <=2 .

In the phase space the equation of motion becomes a system of ODEs:

x'=v

v'=-(α^2 - k Cos[t]) Sin[x] where (x,v) \in S^1 x R .

Points (0,0) and (Pi,0) are the equilibria.

But the suspension point oscillates 2Pi-periodically in time, due to the external force which is 2Pi-periodic, hence it results more convenient to work in the Quotient Extended Phase Space where the orbits traced by the equilibria are given by t-->(0,0,t Mod .2Pi) , t--->(Pi,0,t Mod .2Pi) .

The equation of motion in the Quotient Extended Phase Space S^1 x R x S^1 becomes:

x'=v

v'=-(α^2 - k Cos[t]) Sin[x]

t'=1 mod 2Pi .

From the theory, orbits correspond to fixed points of the Pointcaré Map $\psi_{\alpha,k}$ which is actually the Period Shift Map : $\psi_{\alpha,k} = \phi_{2\pi,0}$.

We need to linearize the Period Shift Map at fixed points, but from the theory its eigenvalues are those of the monodromy matrix without 1.

The monodromy matrix can be computed by solving the variational equation at time T=2Pi,

with initial condition *Id* matrix.

I have :
1) computed aside the Variational Equation for both the orbits t-->(0,0,t) and t-->(Pi,0,t) of the equilibria;
2) integrated the VE equation in both cases with Runge Kutta 2;
3) performed the analysis of the stability/instability regions in (α,k).

Stability/Unstability regions for (0,0) fixed point:

SETUP :

In[5]:=
```
XL[{a_, k_}][{x_, v_, t_}] := {v, -(a^2 - k Cos[t]) x , 1} ;
(*vector field of variational equation in extended phase space*)

RK2[ak_, h_][xvt_] := xvt + h XL[ak][xvt + h/2 XL[ak][xvt]];
(*RK2 to approximate solutions of the variational equation*)

sol[ak_, n_, {x0_, v0_}] := Take[Nest[RK2[ak, 2 π/n], {x0, v0, 0}, n], 2];
(*compute solution of variational equation,
given initial data {x0,v0}, at time T=2π, n iterations*)
(*output just (x,v) first two entries*)

PSM[ak_, n_] := {sol[ak, n, {1, 0}] , sol[ak, n, {0, 1}]};
(*Period Shift Map = Pointcaré Map, given initial conditions (1,0), (0,1)*)
```

USE COMPILE FOR SETUP, IN ORDER TO HAVE FASTER COMPUTATIONS :

In[9]:=
```
RK2[{a, k}, h][{x, v, t}]
```

Out[9]=
$$\left\{x + h\left(v + \frac{1}{2} h x \left(-a^2 + k \operatorname{Cos}[t]\right)\right),\ v + h\left(\frac{h v}{2} + x\right)\left(-a^2 + k \operatorname{Cos}\left[\frac{h}{2} + t\right]\right),\ h + t\right\}$$

In[10]:=
```
RK2c = Compile[{{αk, _Real, 1}, {h, _Real}, {xvt, _Real, 1}},
    {xvt[[1]] + h (xvt[[2]] + 1/2 h xvt[[1]] (-αk[[1]]^2 + αk[[2]] Cos[xvt[[3]]])),
     xvt[[2]] + h (h xvt[[2]]/2 + xvt[[1]]) (-αk[[1]]^2 + αk[[2]] Cos[h/2 + xvt[[3]]]),
     h + xvt[[3]]}] ;

(*RK2 already implemented with vector field XL inside, faster computations*)
```

In[11]:=
```
solc = Compile[ {{αk, _Real, 1}, n, {xv, _Real, 1}},
                    Take[  Nest[RK2c[αk, 2π/n, #] &, Append[xv, 0], n]  , 2] ];

(* # & converts the slot # into a input for the function*)
```

In[12]:=
```
PSMc[ak_, n_] := {solc[ak, n, {1, 0}], solc[ak, n, {0, 1}]} ;

(*Period shift map=Pointcaré Map*)
```

STUDY OF LYAPUNOV STABILITY WRTO (a,k) AND PLOT UNSTABLE POINTS :

We need to study the spectrum of the linearization of PSMc at (0,0), i.e $\Psi_{\alpha,k}$'(0,0), made of 2 eigenvalues.
As noticed in the theory, the dynamical system conserves the volume, hence the product of the two eigenvalues is 1, and by the Lyapunov Stability Criterium:
$|Tr(\Psi_{\alpha,k}'(0,0))| <= 2$ if and only if (0,0) Lyapunov Stable ;
$|Tr(\Psi_{\alpha,k}'(0,0))| > 2$ if and only if (0,0) Lyapunov Unstable .

We choose this derivation of the criterium because it doesn't require to explicitely compute the eigenvalues of PSM, but it's enough to compute its trace.
Then plot unstable points, i.e. unstable regions.

In[13]:=
```
αkTr[ak_, n_] := { ak,  Abs[Tr[ PSMc[ak, n]]  ] } ;     (* ((a,k),|Tr|) *)

RandαkTr[n_, a1_, a2_, k1_, k2_] :=
  αkTr[{RandomReal[{a1, a2}], RandomReal[{k1, k2}]}, n ] ;
(* ((a,k),|Tr|) with a,k random *)

critTr[eps_][lst_] :=  lst[[-1]] > 2 + eps  ;
(*criterion |trace| > 2 + eps *)

UnstTr[lst_, eps_] := Select[lst, critTr[eps]] ;
(* select those ((a,k),|Tr|) with |Tr| > 2 + eps , i.e. unstable *)

PlotUnstTr[lst_, eps_, a1_, a2_, k1_, k2_] :=
  ListPlot[ Transpose[UnstTr[lst, eps]][[1]] ,
   PlotRange → {{a1, a2}, {k1, k2}} , AxesLabel → {"α", "k"} ]  ;

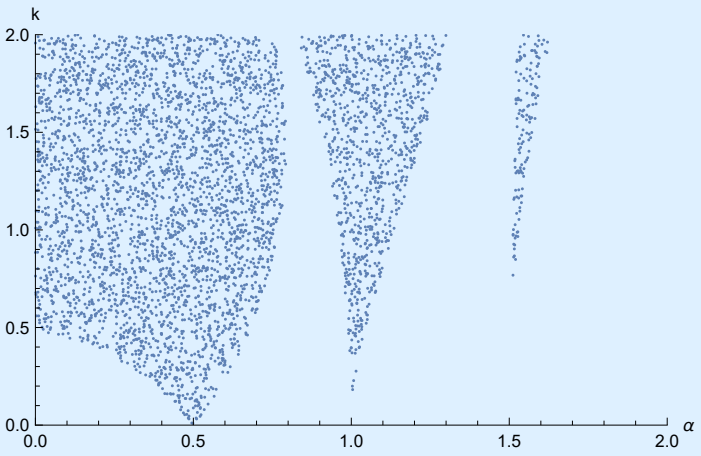
(*plot unstable points (α,k)*)
```

In[18]:=
```
randpoints1 =  Table[RandαkTr[1000, 0, 2, 0, 2], 10000] ;   // Timing
 (*increase number of  points and iterations, for more accuracy*)
```

Out[18]=
```
{10.5625, Null}
```

In[19]:=
```
PlotUnstTr[randpoints1, 0.0001, 0, 2, 0, 2]
```

Out[19]=

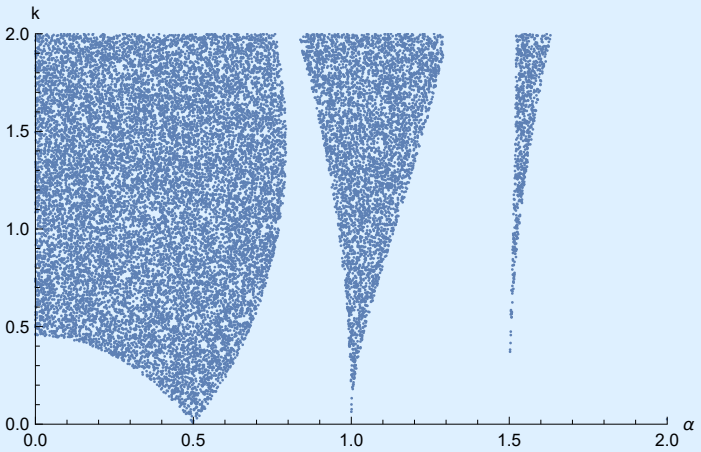


In[20]:=
```
randpoints2 =  Table[RandαkTr[1000, 0, 2, 0, 2], 50000] ;    //
 Timing      (*increase number of points*)
```

Out[20]=
```
{55.875, Null}
```

In[21]:=
```
PlotUnstTr[randpoints2, 0.000001, 0, 2, 0, 2]
(*UNSTABILITY GLOBAL PORTRAIT*)
```

Out[21]=



This plot represents a global portrait of the instability regions, i.e. for which points (α,k) the equilibrium (0,0) is Lyapunov unstable.
While the regions with no points are the stability zones.

The instability regions are tongues (existence proved).

Notice that for k=0 , just for some specific α the equilibria is unstable, where the tongues intersect the horizontal axis .
From the theory we know that this can happen only at $|Tr|=2$ iff $\alpha = n/2$, where n is any integer .
These points are those at stability boundary, where the system transitions from stability to instability regions and viceversa.

If k=0 , more likely (0,0) is stable, but we can force α ,which depends on the frequency of the external force, to make (0,0) unstable.

At $(k,\alpha)=(0, n/2)$ for any positive integer n, the instability tongues end with a cusp point.
In the following I check numerically that at every multiple of 0.5 we have an instability tongue:

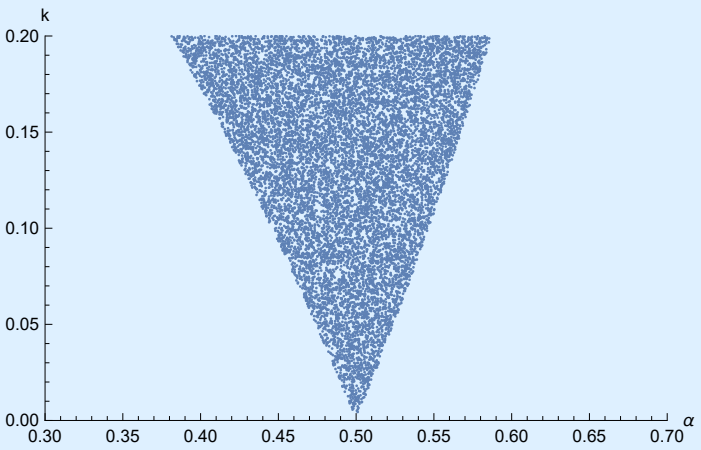
In[22]:=
```
randpoints3 = Table[RandαkTr[1000, 0.3, 0.7, 0., 0.2], 50000] ;  //
 Timing     (*(k,α)=(0, 1/2), n=1*)
```

Out[22]=

```
{53.1094, Null}
```

In[23]:=
```
PlotUnstTr[randpoints3, 0.0001, 0.3, 0.7, 0., 0.2]
```

Out[23]=



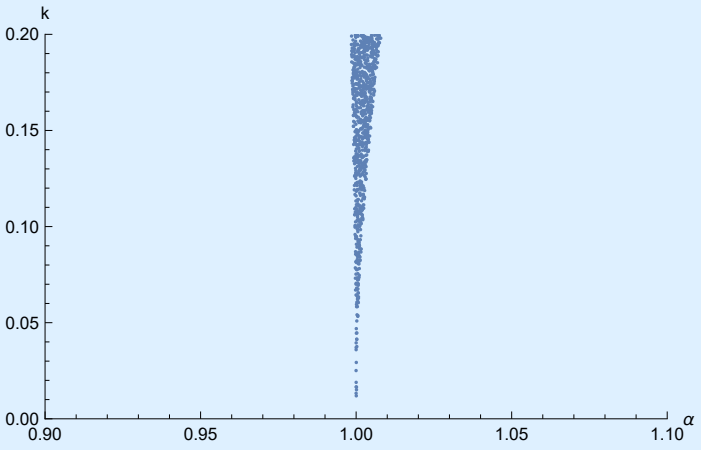
In[24]:=
```
randpoints4 = Table[RandαkTr[1000, 0.9, 1.1, 0., 0.2], 50000] ;  //
 Timing     (*(k,α)=(0, 2/2),n=2*)
```

Out[24]=

```
{53.5, Null}
```

In[25]:=
```
PlotUnstTr[randpoints4, 0.00000001, 0.9, 1.1, 0., 0.2]
```

Out[25]=



Stability/Unstability regions for (Pi,0) fixed point :

Adapt the previous code, it changes just the orbit t-->(Pi,0,t) of the equilibrium , and hence the variational equation.

In[26]:=
```
XLp[{a_, k_}][{x_, v_, t_}] := {v, (a^2 - k Cos[t]) x , 1} ;
(*vector field of variational equation for (Pi,0) equilibria*)
```

In[27]:=
```
RK2cp = Compile[{{αk, _Real, 1}, {h, _Real}, {xvt, _Real, 1}},
    {xvt[[1]] + h (xvt[[2]] + 1/2 h xvt[[1]] (αk[[1]]^2 - αk[[2]] Cos[xvt[[3]]])),
     xvt[[2]] + h (h xvt[[2]]/2 + xvt[[1]]) (αk[[1]]^2 - αk[[2]] Cos[h/2 + xvt[[3]]]),
      h + xvt[[3]]}] ;

solcp = Compile[ {{αk, _Real, 1}, n, {xv, _Real, 1}},
                    Take[  Nest[RK2cp[αk, 2 π/n, #] &, Append[xv, 0], n]  , 2] ];


PSMcp[ak_, n_] :=
  {solcp[ak, n, {1, 0}], solcp[ak, n, {0, 1}]} ;
(*SETUP with Compile*)
```

In[30]:=
```
αkTrp[ak_, n_] := { ak,  Abs[Tr[ PSMcp[ak, n]]  ] } ;     (* ((α,k),|Tr|) *)

RandαkTrp[n_, a1_, a2_, k1_, k2_] :=
  αkTrp[{RandomReal[{a1, a2}], RandomReal[{k1, k2}]}, n ] ;
(* ((α,k),|Tr|) with α,k random *)

critTr[eps_][lst_] := lst[[-1]] > 2 + eps  ;
(*criterion |trace| > 2 + eps *)

UnstTr[lst_, eps_] := Select[lst, critTr[eps]] ;
(* select those ((α,k),|Tr|) with |Tr| > 2 + eps , i.e. unstable *)

PlotUnstTr[lst_, eps_, a1_, a2_, k1_, k2_] :=
  ListPlot[ Transpose[UnstTr[lst, eps]][[1]] ,
   PlotRange → {{a1, a2}, {k1, k2}} , AxesLabel → {"α", "k"} ]  ;

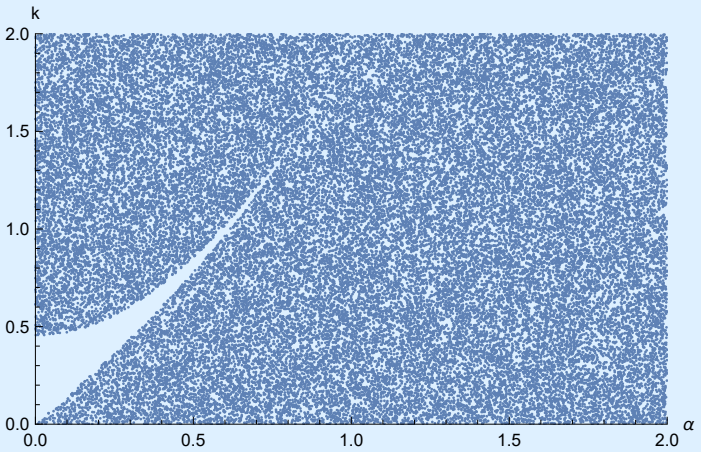

(* PLOT UNSTABLE POINTS (α,k) *)
```

In[35]:=
```
randpoints6 = Table[RandαkTrp[1000, 0, 2, 0, 2], 50000];  // Timing
```

Out[35]=
```
{52.4531, Null}
```

In[36]:=
```
PlotUnstTr[randpoints6, 0.0001, 0, 2, 0, 2]
(*GLOBAL PORTRAIT OF UNSTABLE POINTS*)
```

Out[36]=



The equilibria (Pi,0) is more likely unstable, but we can notice that there exist some (α,k) for which we have stability .
Furthermore, the number of avaiable α's decreases as α increases, e.g. the no-points strip becomes even more thinner as α increases.